

Slides 6.7: Quick Bread Science Lab

Slide outline for Lesson 6.7. Nineteen slides across two days: Day 1 slides 1 to 10, Day 2 slides 11 to 19. Slide 6 is the treatment table and it stays on the board all of Day 2. *[Sal: photograph the four muffins in a row at the end of Day 2. That photo is the review slide for items 9 and 10 on the Cooking Methods Check.]*

DAY 1

Slide 1: Quick Bread Science Lab

- Unit 6, Lesson 6.7, Day 1
- Topic 6.2 Food Preparation Methods and Science
- Four batches, one difference
- This is an experiment, not a recipe Image: four muffins in a row, visibly different heights, cut in half.

Slide 2: Do now

- A muffin and a loaf of bakery bread both rise
- Do they rise the same way?
- Write yes or no and one guess about what makes each one rise Notes: Take four answers, correct none. A student who says “yeast” for the bread has half the concept already; give them the Batch E stretch.

Slide 3: Watch this

Image: a clear cup with baking soda, and a hand pouring vinegar into it, foam climbing over the rim onto a tray. Notes: This is the hook. Do it live, with no warning. Then: “That gas is what lifts a muffin. Today you figure out exactly which ingredient makes it.”

Slide 4: How leavening works

- Baking powder and baking soda both make carbon dioxide gas in a wet batter
- The gas makes bubbles
- The bubbles get trapped and push the batter up
- The heat sets the batter around them, so it stays up after it cools Image: a cross-section drawing of batter with bubbles rising through it.

Slide 5: The difference between the two

- **Baking powder** already has its own acid mixed in. Water and heat are enough
- **Baking soda** has no acid of its own. It needs an acid in the recipe
- Acids in a real kitchen: buttermilk, yogurt, vinegar, lemon juice, brown sugar, cocoa
- No acid means the soda mostly sits there and leaves a soapy taste Notes: Students mark which one needs the acid.

Slide 6: The four batches

Batch	Treatment
A (control)	1 tsp baking powder
B	none at all
C	2 tsp baking powder
D	1 tsp baking soda, no acid

Notes: This slide stays on the board for all of Day 2. Everything else in all four batches is identical.

Slide 7: The four-cup fizz test

- Cup 1: baking soda plus water
- Cup 2: baking soda plus vinegar
- Cup 3: baking powder plus water
- Cup 4: baking powder plus vinegar Notes: Run all four in order and call the result out loud each time. Then ask the class to state the rule from the evidence, not from the slide.

Slide 8: Variable and constants

- **Independent variable:** the one thing we change on purpose. The leavening
- **Dependent variable:** what we measure. Height, crumb, taste
- **Constants:** flour, sugar, salt, milk, oil, egg, batter per liner, oven temperature, bake time, same oven, same shelf
- The control batch is the ruler. Without it, the other three mean nothing Notes: *[Sal: ask the science teacher which words they use for independent and dependent variable and use theirs instead of these. It is free and it makes both classes stronger.]*

Slide 9: Predict

- One prediction per batch
- Rank the four by expected height
- Plus one sentence: what will Batch D taste like, and why? Notes: Predictions are written before any batch exists, so they cannot be adjusted after. Collect nothing; the sheet comes back tomorrow.

Slide 10: Tomorrow

- Roles on your card
- Four bowls, labeled A, B, C, D
- **I dispense the leavening, not you**
- Exit card: the one thing we are changing is . **Batch** will rise the highest because ____ Notes: Say the leavening rule out loud today so it is not a surprise tomorrow. One student adding baking powder to Batch B ends the experiment.

DAY 2

Slide 11: Bake, Measure, Conclude

- Unit 6, Lesson 6.7, Day 2
- Four bowls, one difference, no cheating
- Whatever comes out of that oven is the answer Image: four labeled mixing bowls in a row on a station counter.

Slide 12: Do now

- Read your prediction for Batch B
- Do you still believe it? Yes or no, one sentence Notes: Thirty seconds. Then: “Whatever comes out of that oven is the answer, and it is not up for a vote.”

Slide 13: The six-step routine

- Hair back
- Sleeves up
- Wash hands
- Sanitize surfaces
- Clean as you go
- Station check before dismissal Notes: Same six steps, same order, every lab all year. Start the timer when every station is standing with hands empty.

Slide 14: Mix, in this order

- Dry ingredients measured and **leveled** into the labeled bowl
- I come around with the leavening
- Wet added: milk, oil, egg
- Stir about **fifteen strokes and stop**. Lumps are fine
- 1/4 cup of batter into each labeled liner Notes: Overmixing makes tough batter and it confounds the test. Say “fifteen strokes, lumps are fine” at every station, once.

Slide 15: The flowchart

Minute: 0 5 10 13 20 25 28 35 40

Mix: |=====four batches=====|

Bake: |===13 min===|

Measure: |before| |after, crumb, taste|

Clean: |as you go |ZONES|

Notes: Measure the before height while the batches wait. I load and unload the oven; nobody else touches it.

Slide 16: What to measure

- Height, with the ruler, before and after
- Crumb: split one open. None, few, or many holes

- Taste, if you choose to. Own fork, own plate Image: a ruler held against a split muffin, with the rise marked.

Slide 17: A warning about Batch D

- It will taste soapy or metallic to most people
- Nothing is wrong with you
- That taste **is** the evidence Notes: Say this before anyone tastes. Students who are not warned decide something is wrong with them or with the food.

Slide 18: Conclusion frame

- The data showed ____
- My hypothesis was supported / not supported / partly supported, because ____
- My evidence is ____
- One thing that could have gone wrong is ____ Notes: Take two conclusions out loud. Batch C often rises highest and then collapses; that is the best data in the room and not a mistake.

Slide 19: Station check

- Oven off, tin on the trivet
- Four bowls washed, dried, put away per the map
- Counters wiped and sanitized, sink empty
- Floor swept, trash out
- Leavening jars back on my table Notes: When I sign a station, the Station Manager reads the station's one-sentence conclusion out loud. Lab sheets collected at the door. Keep one muffin from each batch under wrap for the review.